

XM-1,2: The "Rock" and "Roll" of satellites.

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Brief Introduction

- XM-1 & XM-2 nicknamed "Rock n Roll"
- Boeing's HSS-702 satellite bus
- GEO, 85° and 115° W
- Commercial service began September 25, 2001
- Both satellites retired by 2006 :)



Launch



XM-1 launched 5/8/2001, 6:10:29 PM EST

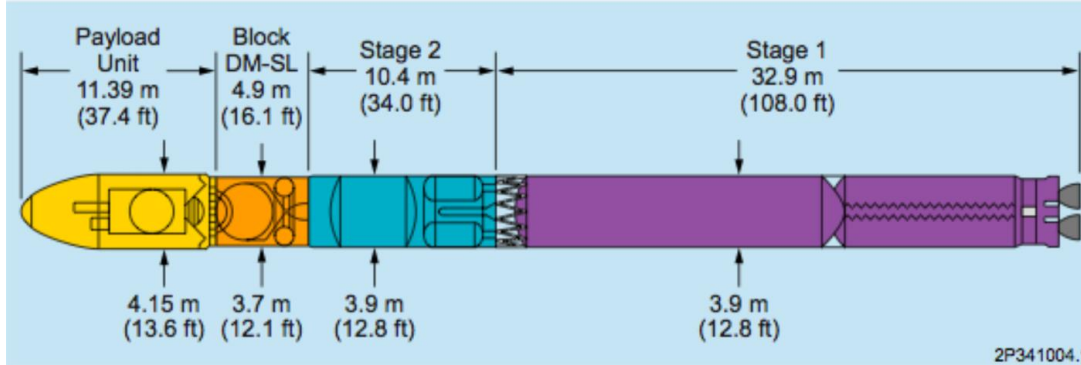
XM-2 launched 3/18/2001, 5:33:29 PM EST

Sea Launch – Ukrainian-based; designed/operated launchpad and transport

- One of the first companies to offer launching at the equator (154° W)
- Oversaw 32 successful launches

Equator launch means less power to launch the same mass into GTO

Three Phase GTO



Each satellite = 1200kg payload -> Zenit-3SL ELV

Liquid oxygen and kerosene fuel

Stages:

1. RD-171 for liftoff
2. RD-120 for thrust; RD-8 for direction
3. Block DM-SL for LEO park, then GTO

Satellite detaches and circularizes orbit into GEO using Xenon Ion Propulsion System

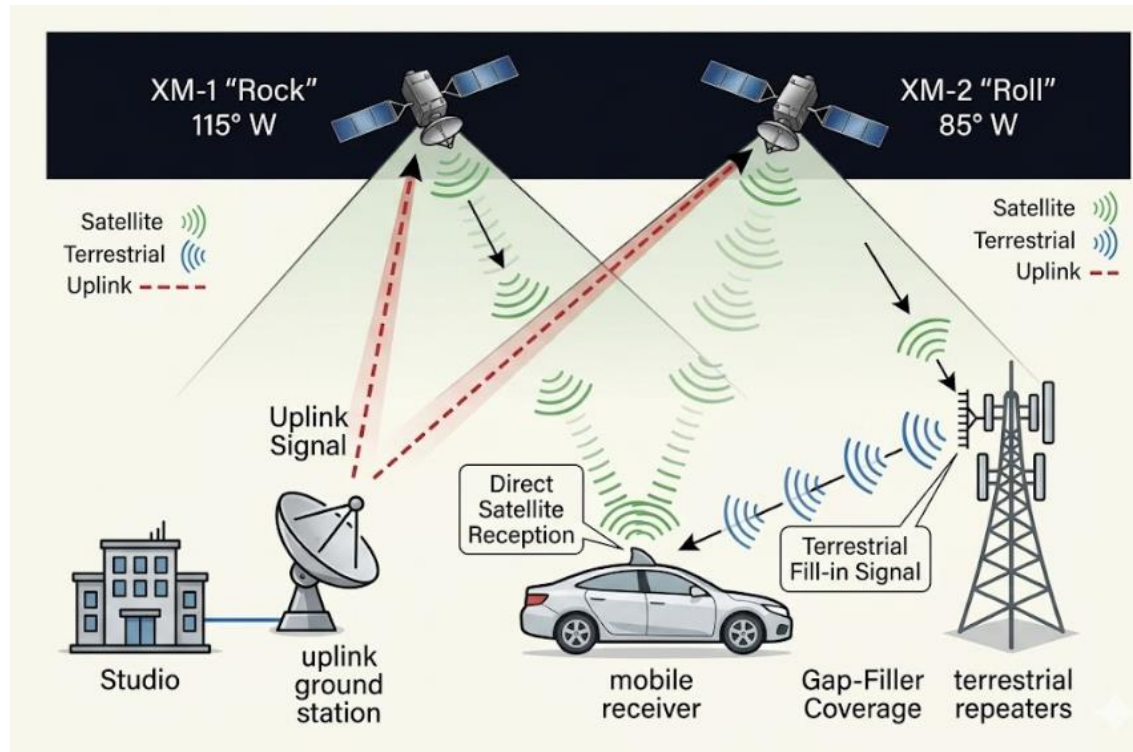


Communication System: Service

- XM provided radio services all over the continental US, with adjacent parts of Canada and Mexico, without using FM/AM.
- Service included ~100 audio channels, with 8kbps to 64kbps data rate each.
- Provided music primarily, along with news, sports and entertainment.
- Car and other mobile radio systems targeted as primary consumers.



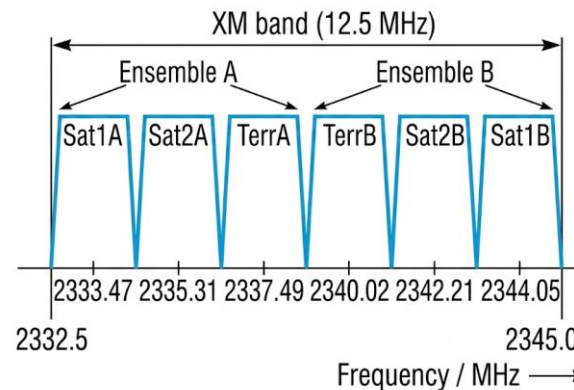
Communication System: Architecture



- Audio channels sourced contents from XM studios, which had facilities in Washington, D.C and New York.
- Nearby ground stations fed uplink data into Rock and Roll.
- The satellites were bent-pipe systems which both transmitted the same (and the whole of) service in downlink.
- Had a terrestrial repeater system at ~60 urban markets.

Communication System: Spectrum Usage

- The uplink from ground stations used X-band frequencies 7.055 GHz to 7.075 GHz. Same feed was transmitted into both satellites.
- The downlink, used the FCC allotted, Satellite Digital Audio Radio Service(SDARS) band from 2332.5 MHz to 2345 MHz.
- The 12.5 MHz total band is split into two ensembles, of 6 * ~2MHz channels.
- ~50 audio channels were transmitted in one ensemble simultaneously, from ~2MHz links of two satellites and the repeater.



<https://www.rohde-schwarz.com/ca/technologies/satellite-broadcast/xm-satellite/>

Comms. System: Coding and Modulation

- In the bent-pipe architecture, the $\sim 2\text{MHz}$ downlink channels is the bottleneck.
- The data stream used 1.886 MHz channels wide, assuming guard band in the remaining.
- XM used QPSK modulation at $1.64\text{ Million sym/s}$ which counts or 3.28 Mbps bit rate.
- 1.64 Msps in 1.886 MHz , lets us assume ~ 0.15 roll-off factor in their Nyquist pulse shaping.
- The documented data rate is over 2 Mbps . Aligning it with the 3.28 Mbps bit rate gives us a ~ 0.64 ratio.
- We assumed a $(2/3)$ FEC coding of 0.67 coding rate, with the rest ~ 0.02 possibly attributed to control overhead.

Uplink Budgeting

Parameter	Value
Uplink antenna gain (dB)	52.1 dB
RF power (dBW)	34.8 dBW
Calculated uplink EIRP (no backoff)	86.9 dBW
Documented uplink EIRP after backoff	70 dBW
Implied uplink backoff	16.9 dB
GEO range	36,000 km
Total uplink loss (pathloss + misc)	205 dB
Satellite uplink receive antenna gain	35.1 dB
Carrier power at satellite input	-99.9 dBW
Uplink system noise temperature	400 K
Uplink noise bandwidth	1.64 MHz
Uplink noise power	-140.5 dBW
Uplink C/N	40.6 dB

Downlink Budgeting

Parameter	Value
Downlink EIRP (example SiriusXM sat)	71 dBW
Downlink path loss (FSPL)	191.2 dB
Atmospheric loss	0.5 dB
Car receive antenna gain	3.7 dB
Carrier power at car receiver	-117 dBW
Downlink system noise temperature	158 K (22 dBK)
Downlink noise bandwidth (two ensembles)	4.5 MHz
Downlink noise power	-140.1 dBW
Downlink C/N	23.1 dB
Overall C/N (dB, cascaded UL+DL)	≈ 22.8 dB

Assuming similar C/N requirements for (2/3) QPSK as (1/2) QPSK, ~6 dB, we get a link margin of ~16.8 dB.

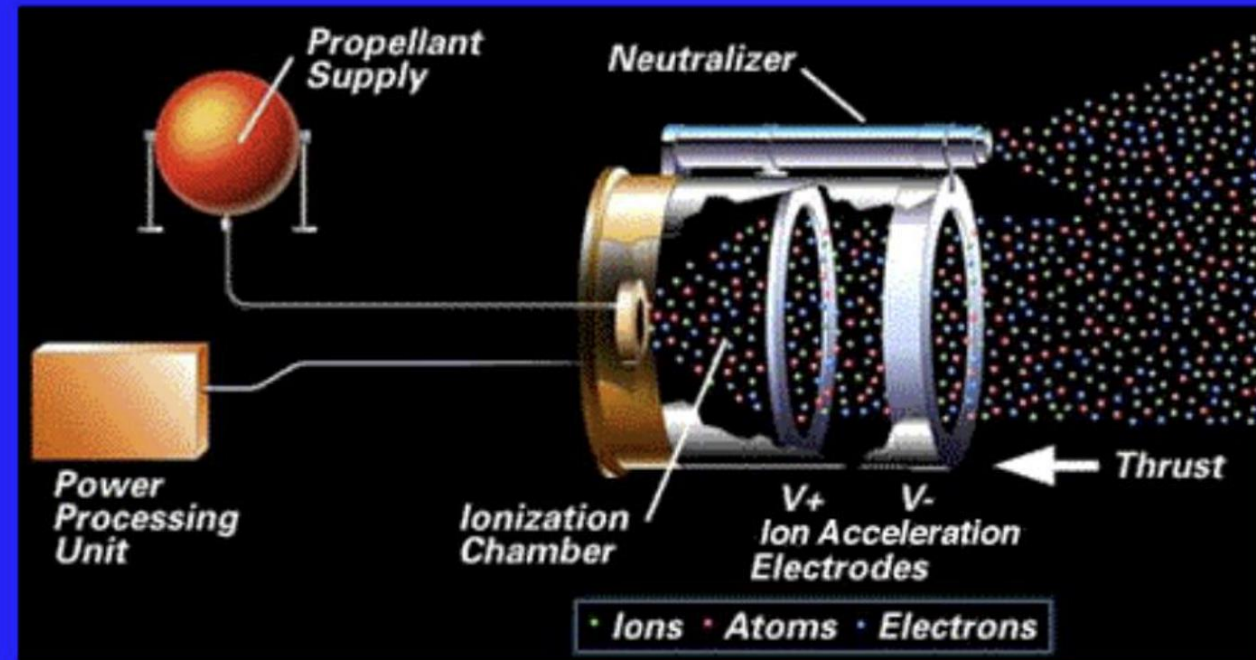
Repeaters:

- Tall buildings and closed spaces create an "urban canyon" situation.
- These environments create a multipath environment that results in poor performance in our ~2MHz wide band link, possibly from frequency selective fading.
- XM uses terrestrial repeater towers for big cities in such case.
- They use coded OFDM, essentially making the data transmissions narrowband and immune to frequency selective fading.
- OFDM is a frequency division multiplexing where subcarriers are orthogonal to each other. XM used (5/9) FEC coding on top of that.

AOCS

- xenon ion propulsion system (XIPS) for on orbit maneuverings
- Four 25-cm XIPS thrusters
- Needing only 5 kg of fuel per year

Xenon Ion Propulsion Systems (XIPS)



Power System

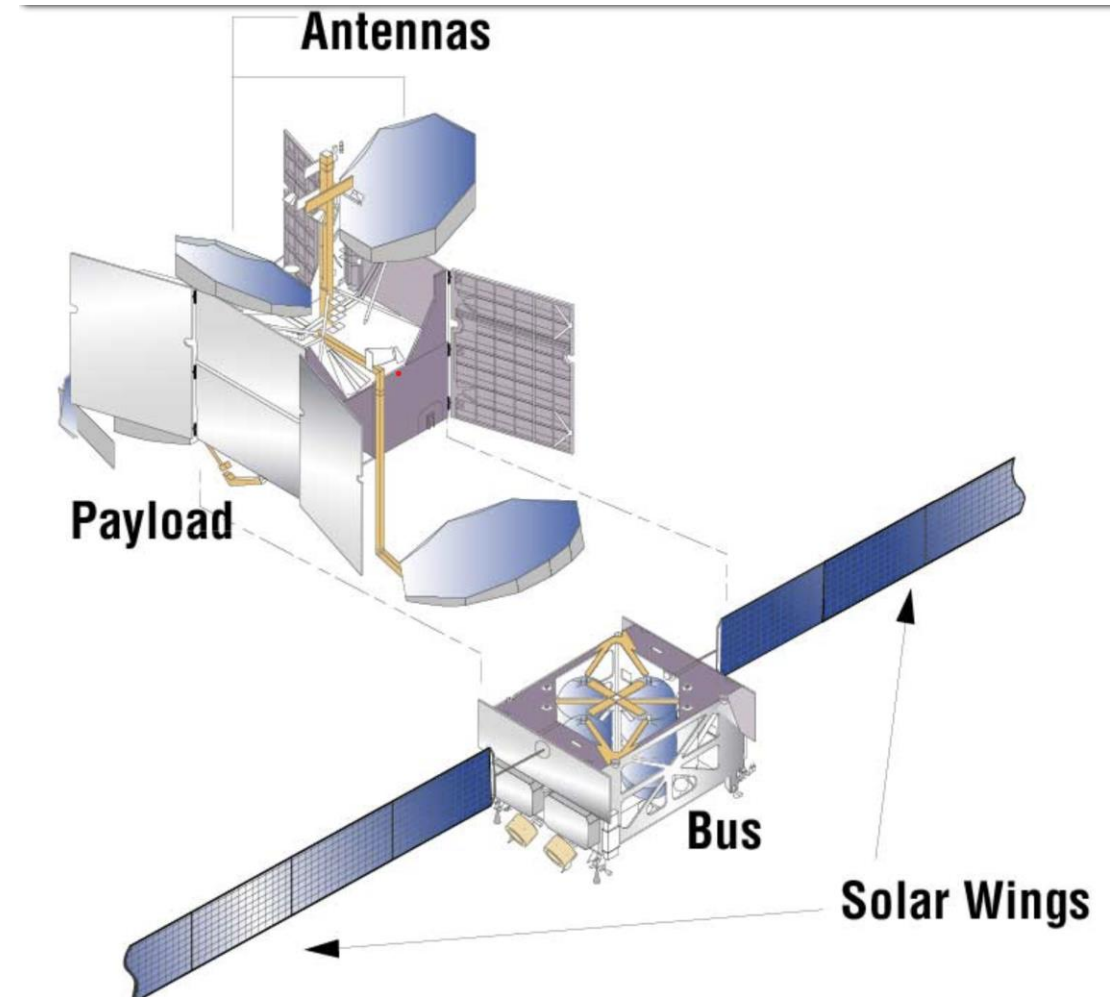
16 228 Watt traveling wave tube amplifiers (TWTAs)

Solar cells provided 18kW BoL, 15kW at 15 years EoL

Two Solar Wings each with 5 solar cells

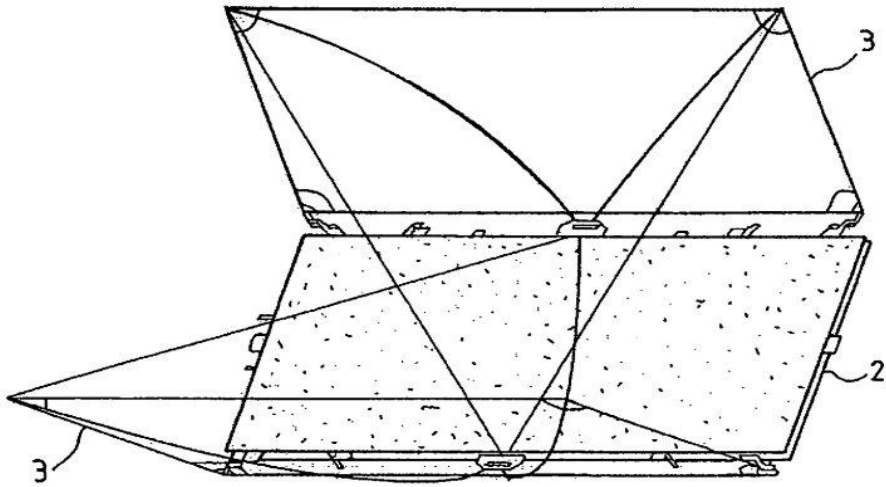
Power System used dual-junction Gallium Arsenide Solar Cells average efficiency of 21%

Included deployable concentrator reflectors increased incident energy by 1.8 times and EoL power 1.4-1.5 times



Boeing: Boeing 702HP Fleet

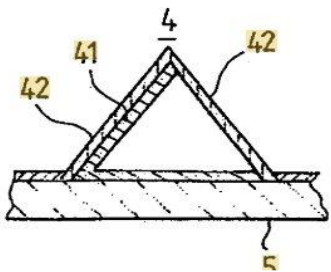
FIG_1



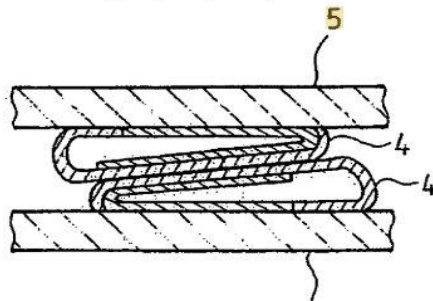
Design Failure: Concentrator Fogging

- Concentrator Reflectors had issues with fogging reducing the amount of available power
- Reduced mission lifespan from ~15 years to 6 years
- Flaw was noticed late on September 26, 2001
- XM Satellite Radio Holding Inc. Reported a 31% stock price drop on, September 27, 2001
- Was a design flaw of the Boeing 702 Bus, meaning 4 other satellites had the same issue
- Insurance companies had to pay out ~\$875 million, with lawsuits ongoing past 2009.

FIG_2a



FIG_2b



Why did this design failure occur?

- Characterization Report shows temperature and radiation lifespan testing
- Concentrator designs had been used before, most notable Deep Space 1, used SCARLETT
- No deployment errors mentioned for solar arms or concentrators

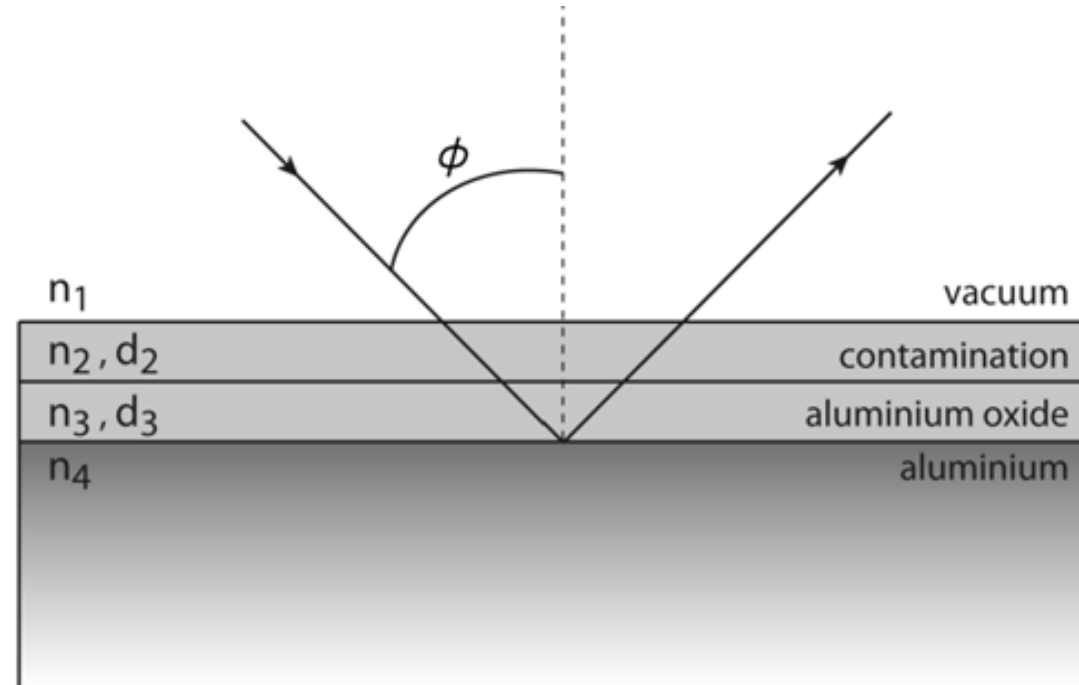


Test	Range/Duration	Result
Optical Characterization		
Solar absorptance, α	integrated over 350-950 nm	$\alpha = 0.12 \pm 0.02$
Normal emissivity	4.8 – 26.2 μm	$\varepsilon = 0.04$
BRDF	measured at 0° and 60°	as measured
Thermal cycling	1856 cycles, –180° to 221°C	no degradation in α , BRDF
Surface Radiation	VDA/CP1 interface dose	no degradation in α , BRDF
Humidity	Six months at 55% RH	no degradation in α , BRDF
Abrasion	Simulated launch environment	<0.1% effect on power
Effect of creases	photometric measurements of reflected light	Negligible effect on power
Mechanical Characterization		
Tensile strength at temperature	–196° C to +200° C	Margin of factor of 60
Strength after thermal cycling	1856 cycles, –180° to 221°C	Margin of factor of 40
Strength after irradiation	7 GRad dose	Margin of factor of 20
Strength after irradiation and thermal cycling at worst-case temperature	7 Grad dose, 1856 cycles, tested at –196° C	Margin of factor of >4
Grounding	Pulsed and steady-state currents at BOL and EOL	Met maximum resistance requirements
CTE/CME/Creep		Dimensionally stable
High temperature vacuum	+235° C for 7 days, $P < 1\text{e-}6$ torr	Met maximum resistance requirements

Table 1. Concentrator material qualification tests.

Outgassing Effect

- The release of microscopic trapped gasses from materials when exposed to a vacuum
- Deposited molecules gather on LOS surfaces and tend to deposit on colder surfaces
- Increases degradation rate for Optical Components
- Optical Equipment are Especially Sensitive to this effect!



Krijger, J. M., R. Snel, Harten, G. van, J. H. H. Rietjens, & Aben, I. (2014). Mirror contamination in space I: mirror modelling. *Mirror Contamination in Space I: Mirror Modelling*, 7(10), 3387–3398. <https://doi.org/10.5194/amt-7-3387-2014>

Irony

"The main benefits are cost savings from the reduction of solar cells, weight reduction (most designs) and smaller stowed envelope (most designs). **In the commercial world, cost is king.** The newest technologies of higher efficiency solar cells and inexpensive thin film solar cells will dim the benefit of concentrator arrays and is making their future uncertain" -Ray Stribling at the Twenty-Eighth IEEE Photovoltaic Specialists Conference - 2000

Future Development

XM-3 & XM-4 had to be launched earlier than expected

- Took over XM-1 & XM-2, which remain in-orbit as stand-by

Removed concentrator design instead used triple junction Gallium Arsenide Cells (Average Efficiency ~26%)

Commercial success

- 1 million customers in 2003
- 18 million by 2008 (merged with Sirius)

Questions?